Task 02: Service Fingerprinting & Banner Grabbing

1. Service Fingerprinting Using Nmap

I Use Nmap's Version Detection (-sV) and Aggressive Scan (-A) flags to identify service details.

Command

nmap -sV -sS -T4 -p1000 hackthissite.org

➔ *-sV (Version Detection)*

This flag tells Nmap to detect the **exact service and version** running on open ports by sending probes and analyzing responses.
This scan is doing stealthier scan on top 1000 ports of hackthissite.org with aggressive speed

POC

```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ nmap -sV -sS -T4 -P 1000 hackthissite.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 14:44 +0500
Nmap scan report for hackthissite.org (137.74.187.101)
Host is up (0.20s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.100 137.74.187.
rDNS record for 137.74.187.101: 101.187.74.137.in-addr.arpa
Not shown: 997 filtered tcp ports (no-response)
PORT     STATE  SERVICE        VERSION
22/tcp   closed ssh
80/tcp   open   http-proxy     HAProxy http proxy 1.3.1 - 1.9.0
443/tcp  open   ssl/http-proxy HAProxy http proxy 1.3.1 - 1.9.0
Service Info: Device: load balancer

Service detection performed. Please report any incorrect results at https://nm
Nmap done: 2 IP addresses (1 host up) scanned in 80.93 seconds
```

Here Nmap not only shows **open ports** but also the **exact service versions**.

Aggressive Scan Using Nmap

nmap -sV -sS -A -p1000 hackthissite.org

-A Includes

Aggressive scan automatically enables:

- Version detection
- OS detection
- Script scanning
- Traceroute

So, it provides **detailed information about the target system**.

POC

```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ nmap -sV -sS -A -p1000 hackthissite.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 14:52 +0500
Nmap scan report for hackthissite.org (137.74.187.104)
Host is up (0.19s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.101 137.74.187.100 137.74.1

PORT     STATE    SERVICE VERSION
1000/tcp filtered cadlock
Too many fingerprints match this host to give specific OS details
Network Distance: 18 hops

TRACEROUTE (using port 443/tcp)
HOP RTT       ADDRESS
1   4.00 ms   192.168.1.1 (192.168.1.1)
2   6.36 ms   39.43.128.1
3   6.00 ms   10.253.13.54 (10.253.13.54)
4   6.66 ms   10.253.12.18 (10.253.12.18)
5   26.28 ms  10.253.4.38 (10.253.4.38)
6   26.28 ms  10.253.4.4 (10.253.4.4)
7   164.62 ms fra1-lim3-vac-1-firewall.de.eu (178.33.99.161)
8   164.75 ms 37.59.16.56
9   ...
10  164.78 ms be105.lil1-rbx1-sbb1-nc5.fr.eu (178.33.100.160)
11  154.97 ms 37.59.16.12
12  ...
13  154.63 ms be105.lil1-rbx1-sbb1-nc5.fr.eu (178.33.100.160)
14  157.95 ms 37.59.16.20
15  ... 17
18  164.67 ms hackthissite.org (137.74.187.104)

OS and Service detection performed. Please report any incorrect results at https://nmap.or
Nmap done: 1 IP address (1 host up) scanned in 19.81 seconds
```

2.Manual Banner Grabbing

Sometimes automated tools may miss details, so testers manually read the service banner

```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ nc hackthissite.org  80
HTTP/1.0 408 Request Time-out
Cache-Control: no-cache
Connection: close
Content-Type: text/html

<html><body><h1>408 Request Time-out</h1>
Your browser didn't send a complete request in time.
</body></html>
```

SMB Enumeration (If Port 445 Is Open)

As our scan shown no SMB is runnning on the target

If there were any service running on the port 445 we will be doing these step to extract vaulable information of the target

You can enumerate it using **Enum4linux**.

Command

enum4linux hackthissite.org

➔ enum4linux

This tool gathers information from SMB services such as shared folders, user accounts, and system details.

What It Can Discover.

- Shared folders
- User accounts
- Password policies
- Domain information

CONCLUSION

In penetration testing, simply finding an open port is not enough. The real objective is to identify the exact service and version running on that port, because vulnerabilities and exploits are usually specific to particular software versions. By fingerprinting services and grabbing banners, a security tester can map the target system to known CVEs and determine whether an exploitable vulnerability exists.